

Infix to postfix

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- 1) start scanning from left to right
- 2) If(current element is an operand)
Append it to the postfix expression
- 3) Else if (current element is opening bracket '(')
Push it onto the stack
- 4) Else if current element is closing bracket ')'
 - Pop elements from the stack and append them to postfix exp till its corresponding opening brackets does not occur
 - Pop opening brackets from the stack and discard both the brackets
- Else
 - // If(current element is an operator)
 - While(stack is not empty && priority of topmost element >= priority of current element)
 - {
 - Pop element from the stack and append it to postfix expression
 - }
 - Push current element onto the stack.
- 5) Repeat the above steps till end of infix expression
- 6) Pop all the remaining elements from the stack one by one and append them to postfix expression

Infix expression :

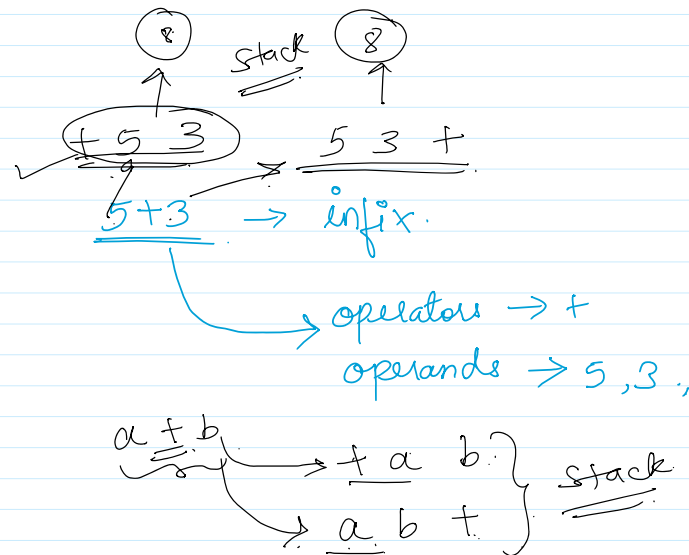
$5+9-4*(8-6/2)+1*(7-3)$

Current element :

Postfix expression: $59+4862/-* -173- \$ +$

stack

top



$5+9-4*(8-6/2)+1*(7-3)$

Handwritten conversion steps for the expression $5+9-4*(8-6/2)+1*(7-3)$:

Step 1: $5+9-4*(8-6/2)+1*(7-3)$ (Initial expression)

Step 2: $5+9-4*(8-6/2)+1*(7-3)$ (Processing the innermost parentheses)

Step 3: $5+9-4*(8-6/2)+1*(7-3)$ (Processing the innermost parentheses)

Step 4: $5+9-4*(8-6/2)+1*(7-3)$ (Processing the innermost parentheses)

Handwritten stack state for the expression $5+9-4*(8-6/2)+1*(7-3)$:

Stack (from top to bottom):

- $($
- $^$
- $*/$
- $+ -$

$$\begin{aligned}
 & 5 + 9 - 4 * \underline{8621-} + \underline{1\$73-} \\
 & 5 + 9 - 4 * \underline{8621-} + \underline{173-\$} \\
 & \underline{5+9-4} \underline{8621-*} + \underline{173-\$} \\
 & \underline{59+} - \underline{48621-*} + \underline{173-\$} \\
 & \underline{59+48621-*} - + \underline{173-\$}
 \end{aligned}$$

$$\boxed{59+48621-* - 173-\$ +}$$

Postfix expression: 59+48621/*-173-\$+